

JAYANT KUREKAR (Full-Stack Software Engineer)

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PROFILE SUMMARY

Full-stack software engineer with 2 years of software development experience with Python/FastAPI, Node.js, and React/TypeScript across AI infrastructure, Web3, and industrial IoT. Built InfrGate, a multi-tenant LLM API gateway with automatic failover and usage metering, and Walnut, an FHE-encrypted lending protocol backed by a Fhenix Foundation grant. 8x hackathon winner across 15+ offline hackathons. Comfortable owning a system end-to-end — from database schema and API design through deployment.

EXPERIENCE

Stealth Startup — Remote

May 2025 – April 2026

Full Stack Engineer

- **Built and deployed AI bots** handling customer-facing interactions, cutting manual response load across inbound workflows.
- **Automated end-to-end WhatsApp workflows** — AI-driven conversations at scale with zero human handoff on routine tasks.
- **Shipped full-stack product features** - REST APIs, React frontend, Python and FastAPI backend — on tight weekly cycles from design to production.

Edunet Foundation × Microsoft

Apr 2025 – May 2025

AI & Cloud Internship Trainee

- **Designed and deployed a production AI system on Azure AI and the OpenAI API**, integrating scikit-learn models from data pipeline to cloud inference endpoint.
- **Applied deep learning and generative AI techniques** under direct mentorship from Microsoft engineers across 40+ hours of structured ML and cloud architecture training.
- **Certified by AICTE** for cloud and AI proficiency, with hands-on work across Azure Cognitive Services, OpenAI embeddings, and scikit-learn pipelines.

PROJECTS

InfrGate — Multi-Tenant LLM API Gateway

[Live](#) | [GitHub](#)

Stack: Python · FastAPI · PostgreSQL (Supabase) · Redis (Upstash) · SQLAlchemy 2.0 · Docker

- **Architected an OpenAI-compatible API gateway** that lets client apps swap LLM providers with zero code changes, while gaining automatic failover, spend caps, and usage metering out of the box.
- Built an async **FastAPI backend with SQLAlchemy and PostgreSQL** as the system of record for tenant API keys, model configuration, and a durable token-usage ledger.
- Implemented **token-bucket rate limiting and sliding-window circuit breakers on Redis**, plus a **SKIP LOCKED-based background worker** for contention-free concurrent job processing.
- Hardened the gateway through targeted testing — closing a **webhook SSRF vector via DNS rebinding and fixing async cancellation handling** — then deployed it as a containerized service on Render, multiplexing across Hugging Face, Gemini, and OpenAI.

Walnut — Confidential On-Chain Lending Protocol [Fhenix Foundation Grant Recipient]

[Live](#) | [GitHub](#)

Stack: Solidity · Fhenix CoFHE SDK · React · TypeScript · Hardhat · Ethers.js

- Built the first **FHE-native confidential lending protocol** on Fhenix — collateral, debt positions, health factors, and liquidation thresholds **stay encrypted end-to-end**, with protocol logic executing directly on ciphertext.
- Redesigned the decryption architecture from a push-callback to a client-pull model during security audits, closing a critical vulnerability class alongside underflow and state-restoration fixes.
- **Designed a unified encrypted vault** with an on-chain risk engine — health-factor computations run entirely on encrypted inputs, keeping user balances encrypted throughout.
- Routed state transitions through a private mempool instead of the public mempool to **reduce MEV extraction and liquidation front-running risk**.

IEMAS — Full-Stack Industrial IoT Energy Monitoring Platform

[GitHub](#)

Stack: FastAPI · Python · SQLAlchemy · Pydantic · PostgreSQL · Supabase · Next.js · React · TypeScript · ESP32 · Modbus RTU/TCP

- **Architected an end-to-end industrial IoT pipeline** — Schneider energy meters (Modbus RTU/TCP, with a realistic simulator for offline testing) feed through ESP32 edge devices into a FastAPI backend, a Supabase-hosted PostgreSQL database, and a Next.js dashboard.
- Built an **async FastAPI backend with SQLAlchemy ORM and Pydantic validation**, handling data ingestion, persistence, and API design.
- Developed a Next.js/React/TypeScript dashboard with **role-based access control** for real-time and historical energy analytics across industrial sites.
- **Integrated an AI chatbot** into the dashboard for natural-language interaction with energy consumption data.

TECHNICAL SKILLS

- **Languages:** Python, Java, JavaScript, TypeScript, SQL, Solidity
- **Frameworks & Libraries:** FastAPI, React, Next.js, Node.js, Express.js, SQLAlchemy, Pydantic, Hardhat, Ethers.js
- **Databases & AI Infrastructure:** PostgreSQL, Redis, Docker, Kubernetes, Supabase, Firebase, Azure AI, AWS basics, Google Cloud Console
- **APIs & Backend:** REST APIs, JWT, LLM API Integration (OpenAI, Gemini), Rate Limiting, Circuit Breakers, Kafka
- **Tools:** Git, GitHub, Postman, Cursor, Claude Code, Hugging Face, Gradio, Google Colab, Render, Upstash

ACHIEVEMENTS

- Winner, Fhenix Buildathon — \$5,000 grant for Walnut, a confidential lending protocol
- Winner, Aleo Buildathon — \$1,800 grant
- Winner, ETHMumbai Hackathon — international Web3 hackathon (ENS Track)
- 8 hackathon wins across 15+ offline hackathons, including MLH, IIT, ETHGlobal, and IIIT events